Control Methods for a Four-Leg NPC Inverter with Balanced and Unbalanced Input Signals

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***Abstract*–Creating resilient and efficient microgrids necessitates the utilization of flexible inverters to fulfill user power requirements. This demand for versatility also gives rise to the necessity for innovative control techniques to govern these emerging adaptable inverters. The following report functions as an initial comparative analysis between several common control strategies, in the context of the emerging demands of modern microgrid**

I. Introduction

The reliability and adaptability of inverters is a major factor in the effectiveness and application of microgrids. This paper addresses some control strategies for inverters with the goal of microgrid applications. The control strategies that follow were tested in simulation (using MATLAB Simulink) under both balanced and unbalanced input conditions. The control strategies were implemented on a 4-leg 3-phase NPC inverter, pictured below in Fig. 1.

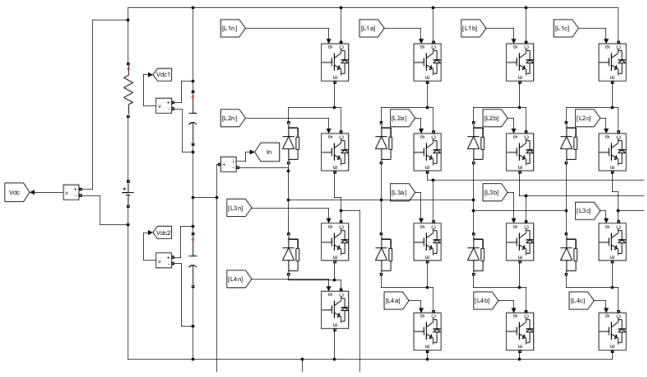


Fig. 1. Four-Leg Three-Phase NPC Inverter Modelled in Simulink

A simple closed-loop control loop was used for current control of the inverter (pictured in Fig. 2). Additionally, the inverter was assumed to be tied to the microgrid.

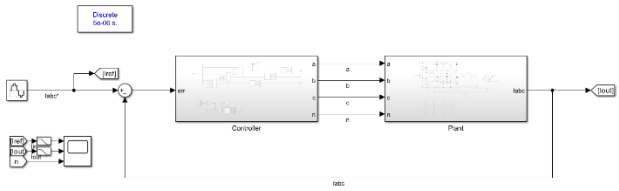


Fig. 2. Inverter Current Control Loop

In total ten different configurations were tested, made up of all possible combinations of the following inverter topologies and control strategies:

Topologies:

1. 3-Leg NPC
2. 4-Leg NPC

Three–Phase Control Strategies:

1. Clarke and Park Control
2. Proportional Resonant Control

Neutral Control Strategies:

1. Three–Phase Injection
2. Fourth Leg Control
3. Unconstrained Fourth Leg

Each of these strategies are described in sections II and III. Although the variational results of each of the combination strategies are shown and discussed in section IV.

II. Three–Phase Control Strategies

A. Clarke and Park Control

The Clarke and Park Transformation is relatively standard in power inverter controls as well as variable frequency drives (VFDs) as it allows a balanced 3-Phase system to be converted into its 0, direct, and quadrature reference frame that can then be easily controlled with a Proportional Integral (PI) Controller, Fig. 3.

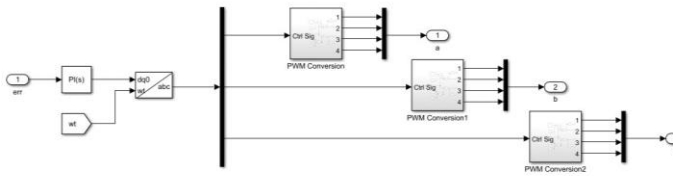


Fig. 3. Park–PI Controller

This controller is effective under balanced input conditions but is not effective under unbalanced input conditions (see section IV for variational simulation results). It should be noted however that a Fortesque transform can be performed before the park transform in order to increase the controller's efficiency under unbalanced input conditions, but this is at the cost of computation time.

B. Proportional Resonant Control

The Proportional Resonant (PR) controller (Fig. 4) works in a similar manner to a PI controller, but instead of acting as a lowpass filter it acts as bandpass filter, thus allowing only the desired frequency of oscillations to be maintained while attenuating the noise.

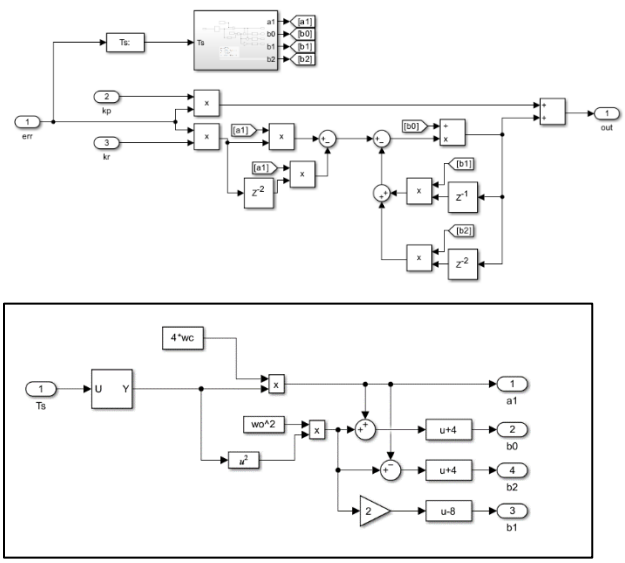


Fig. 4. PR Controller Block Diagram

As hypothesized, this form of controller outperformed the Clarke & Park controller in unbalanced input conditions, due to each leg being controlled separately (shown in Fig. 5).

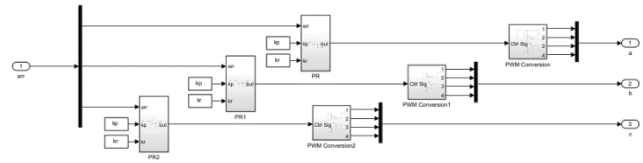


Fig. 5. Three Separate PR Controllers for Each Leg

III. Neutral Control Strategies

A. Three–Phase Injection

Three-phase injection is a method for regulating the neutral point of an NPC inverter. This method is implemented by creating a proportional controller, with the NPC's neutral point voltage, and adding this calculated value to each of the phases to level the and perturbations in the neutral point that may occur overtime. The calculated value from the proportional controller is added to the control loop after the "three-phase control strategy" is implemented. Fig. 6, below, shows the injection control strategy implemented on a PR controller, where the injection takes place between the PR controller and the PWM conversion of the signal.

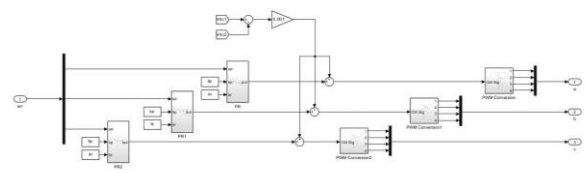


Fig. 6. Three-Phase Injection Control for a PR Controller

It should be noted that this strategy is particularly vulnerable to oversaturation. This is due to the nature of the strategy, in which the "three-phase control strategy" and proportional injection must be tuned separately. Both controllers are then added together, which can result in unpredictable control signal amplitudes, if the neutral point is allowed to get too far out of its reference range.

B. Fourth Leg Control

Fourth Leg Control functions in a similar manner to the *Three-Phase Injection* strategy, using a proportional controller based on voltage measurement of the neutral point. However, this strategy is implemented on a 4-Leg inverter and takes advantage of the 4th leg to regulate the neutral point. The proportional controller is routed into the 4th, converted to PWM, and the neutral point can be regulated (Fig. 7).

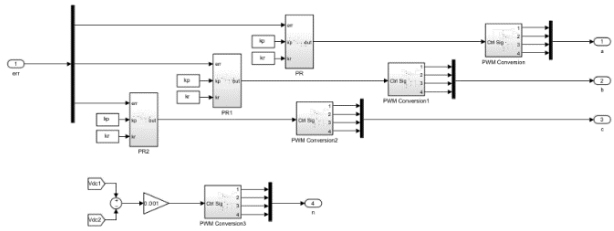


Fig. 7. Fourth Leg Control for a PR Controller

Compared to the *Three-Phase Injection* strategy, the fourth leg control strategy lacks the propensity to oversaturate, as the neutral control is managed by a separate leg. The fourth leg also allows for greater proportional gain without the fear of saturation, when compared to the *Three-Phase Injection* strategy.

IV. Simulation and Results

The following section shows the simulation results of all possible combinations of the strategies mentioned in the previous sections of this paper, with both 3-leg and 4-leg NPC inverters, under both balanced and unbalanced input conditions. The results display the output currents for each leg and the neutral point voltage. Below is the unbalanced input block conditions.

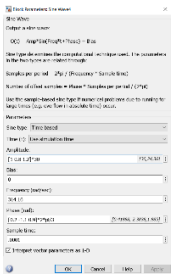
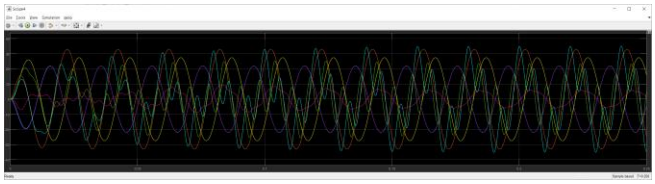


Fig. 8. Unbalanced Input Conditions

A. 3-Leg PR Control

i. Balanced Conditions:

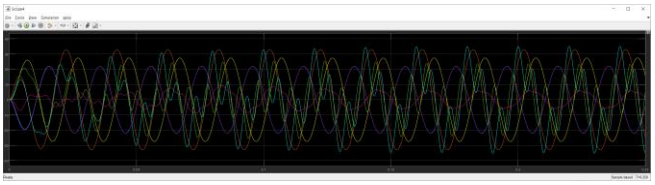
ii. Unbalanced Conditions:



B. 3-Leg Park Control

i. Balanced Conditions:

ii. Unbalanced Conditions:



C. 4-Leg PR Control

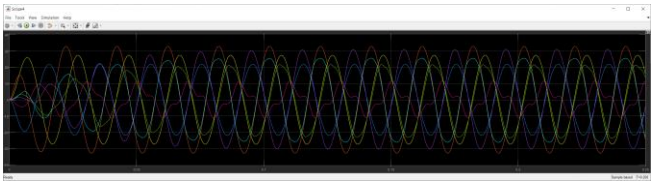
i. Balanced Conditions:

ii. Unbalanced Conditions:

D. 4-Leg Park Control

i. Balanced Conditions:

ii. Unbalanced Conditions:



E. 3-Leg PR Injection

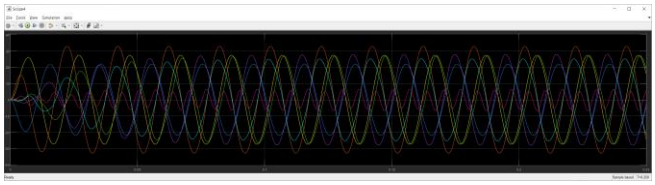
i. Balanced Conditions:

ii. Unbalanced Conditions:

F. 3-Leg Park Injection

i. Balanced Conditions:

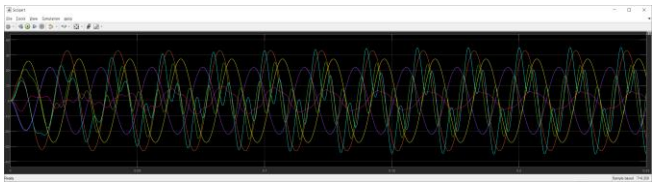
ii. Unbalanced Conditions:



G. 4-Leg PR Injection

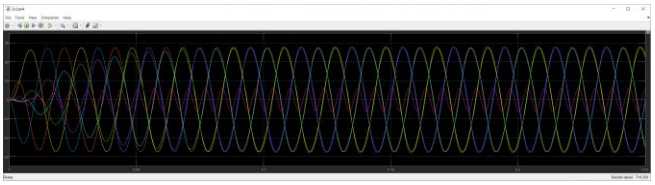
i. Balanced Conditions:

ii. Unbalanced Conditions:

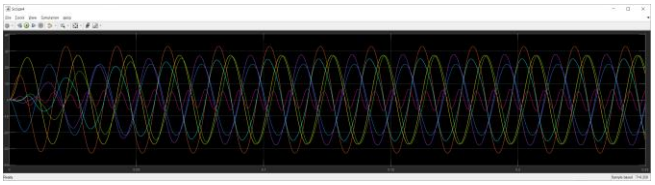


H. 4-Leg Park Injection

i. Balanced Conditions:



ii. Unbalanced Conditions:



I. 4-Leg PR 4th Leg Control

i. Balanced Conditions:

ii. Unbalanced Conditions:

J. 4-Leg Park 4th Leg Control

i. Balanced Conditions:

ii. Unbalanced Conditions:

V. Conclusions and Future Work

From this investigation it was concluded that a PR controller with 4th leg neutral compensation was most suitable for this application (from the tested control methods).

Further work should be done in studying the control strategies under unbalanced input conditions. Dynamic input conditions and HIL simulations should also be analyzed before any useful conclusions can be made. Finally, use of the Fortescue transform in conjunction with the Park transform should be analyzed.

References

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